

NEW PRODUCTS

MAKE IN INDIA

COMPONENTS & CHIPS

Connectivity module for smart management and security

The Swiss semiconductor giant u-blox has introduced SARA-S520M10L, an Internet of Things (IoT) module for cellular and satellite connectivity.



With precise low-power positioning, this module is designed for asset track-

ing, fleet management, maritime transportation, mining, utilities, and smart agriculture applications. It also caters to anti-theft systems, industrial monitoring, control, and safety-critical communication needs. It ensures connectivity even when no terrestrial cellular network is available. It is the most compact multi-mode cellular and satellite IoT module. This module offers seamless connectivity via terrestrial cellular networks and geostationary (GEO) satellites. It maintains pin compatibility with other u-blox cellular-only modules in the SARA form factor. With support for up to four satellite constellations, this module can deliver precise location fixes. It features the latest u-blox UBX-R5 chip's second generation, designed to enable satellite IoT communication in conjunction with LTE-M/NB-IoT cellular standards.

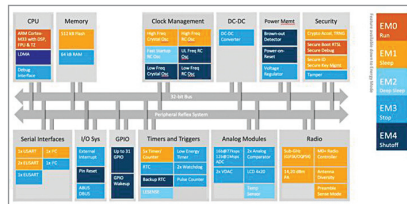
u-blox

<https://www.u-blox.com/en>

Systems-on-chips designed for Amazon Sidewalk

Silicon Labs has announced a new family of systems-on-chips (SoCs)

optimised for Amazon Sidewalk, the SG23 and SG28 SoCs. The SoCs are developed with a versatile portfolio encompassing different device families and variations suited for specific components, including the BG series for Bluetooth SoCs and the ZG series for Z-Wave SoCs. The SG series enhances Amazon Sidewalk devices by integrating Bluetooth Low Energy (BLE) with extended-range protocols



like FSK or CSS, making it ideal for battery-operated sensors due to its low power consumption in both active and sleep modes. Additionally, the device ensures premium IoT security through its specialised security core that endorses secure vault high, protecting the device and wireless data from various potential threats. The SoC enables manufacturers to simplify device design and cut costs by merging the primary radios for Sidewalk devices into one unit. Meanwhile, the SG23 prioritises robust security and a potent sub-GHz connection, suitable for long-distance endpoint devices.

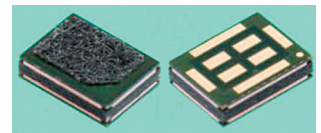
Silicon Labs

<https://www.silabs.com>

Waterproof MEMS microphone carries IPX7 rating

CUI Devices' audio group has launched the CMM-3424DT-26165-TR micro-electromechanical systems

(MEMS) microphone designed for use in challenging and demanding environments. This microphone has been engineered to meet the stringent requirements of harsh conditions, earning an impressive IPX7 waterproof rating. This rating ensures the microphone's resilience against moisture



and liquid ingress, making it a

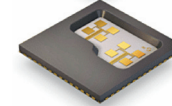
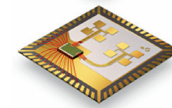
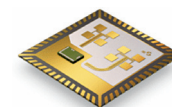
reliable choice for applications where environmental durability is paramount. The waterproof microphone has digital pulse-density modulation (PDM) output, which enhances audio signal processing capabilities. It is equipped with a top port location, optimising its performance for specific applications. This MEMS microphone offers designers great flexibility with its wide operating frequency range, spanning from 100 to 10,000Hz, and a generous operating temperature range extending from -40 to 100°C.

CUI Devices

<https://www.cuidevices.com>

Commercial 240GHz radar transceiver

Indie Semiconductor has launched the world's first commercial fully integrated 240GHz radar front-end (RFE)



silicon transceiver TRA240091. The new system offers an unprecedented sensing resolution and system performance, setting it apart in the field. This fully integrated solution combines all transmitter,